

# Cross Sections of Real Life: Conics

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## 1 Introduction

Little Timmy throws a baseball to his friend 10 feet away. It would be standard to model this trajectory with a parabola. We can say this by making the simplification that the only accelerating force that acts on the ball is Earth's gravity.

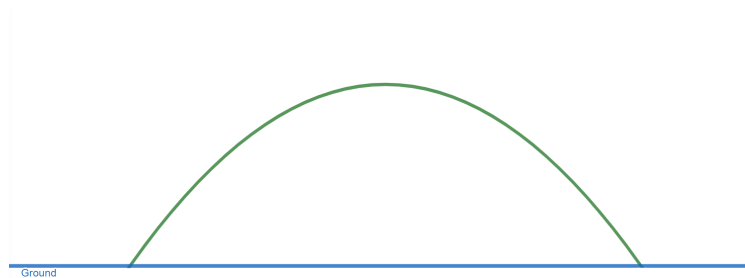


Figure 1: The baseball's path

Now, this works fine when we only want to see the ball's path from Timmy to his friend. But our parabolic path will keep going. Ignoring the physical body of the Earth, this is what our parabola models for the path of the ball after Timmy throws it:

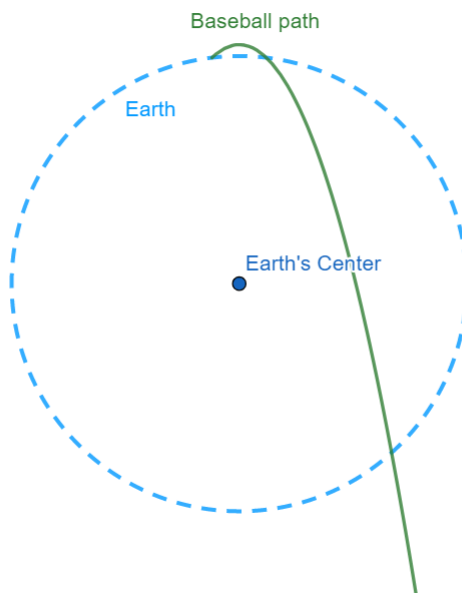


Figure 2: Baseball's path (not to scale)

Yeah...Something's off. This baseball is getting launched out of Earth's orbit into space, somehow reaching escape velocity, with only the Earth's gravity accelerating it?? Of course, that isn't possible. Earth's gravity pulls mass towards the center, so this baseball can't just go flying in this parabolic path.

Here we find the mistake that led to this issue: We take the Earth's gravity to be a constant accelerating force downwards; however if we look from this perspective, the accelerating force of gravity will be changing direction and magnitude depending on how far the ball is from the center. The parabolic approximation is good enough when the distance from the Earth's center barely changes, like we have in Timmy's example, but that does not apply to the general case.

Consider a satellite in orbit. We know that the reason it revolves around Earth is due to Earth's gravitational pull, and it revolves *around* the earth, in a somewhat circular path. This gives us a hint as to what should be happening to the ball: It won't go in the same direction forever, instead the ball will be pulled towards the center of the Earth. Now, let's ignore the other problems presented by the physical body of the Earth and just say that Little Timmy throws his baseball high enough to get into Earth's orbit.

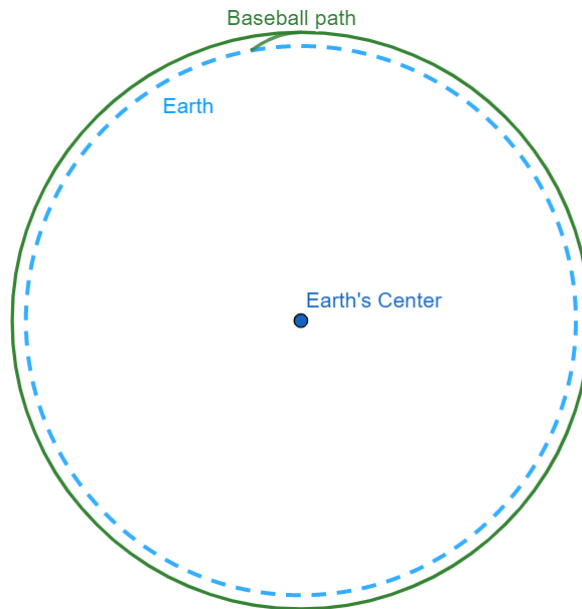


Figure 3: Baseball's orbit (not to scale)

Now, even though Timmy is still just throwing the ball, the path is now somewhat circular. *Somewhat*. The Earth isn't perfectly spherical, however, as the circumferences around the poles and around the equator are different. So, the ball's orbit won't exactly be circular, but **elliptical**. In fact, the path in Figure 2 should be elliptical as well.

We like to simplify curved surfaces into circular pieces because dealing with circles is a lot easier than dealing with other curves. A circle, with its infinite symmetries, represents a perfect, ideal object. This perfection is near impossible to recreate in real life, but it is often useful if we want to approximate values.

So what gives? Parabolic paths are actually elliptical? Ellipses are basically circles? Parabolas are actually circles? These objects seem to have little in common, but yes, they are all connected. Parabolas, circles, and everything in between can all be classified as **conic sections**.

## 2 Definitions

### 2.1 Sections of a Cone

The most direct definition of conic sections is the set of curves formed by the intersection of a plane and a double cone, or two right cones on top of one another. This “double cone” can also be seen as the solid of revolution of a line  $y = mx + b$  about the  $y$ -axis.



Figure 4:  $z^2 = x^2 + y^2$  (Note these cones have no “base.”)

We can obtain all four types of conic sections depending on the plane that intersects with the double cone. The cross section is a circle if the plane is parallel to a base of the cones. An ellipse is formed when the plane intersects one of the cones four times. A circle is sometimes considered a special case of an ellipse. A parabola is formed when the slope of the plane is equal to the slope of a line containing the sides of the cones. Finally, a hyperbola is formed when the plane intersects each cone two times.

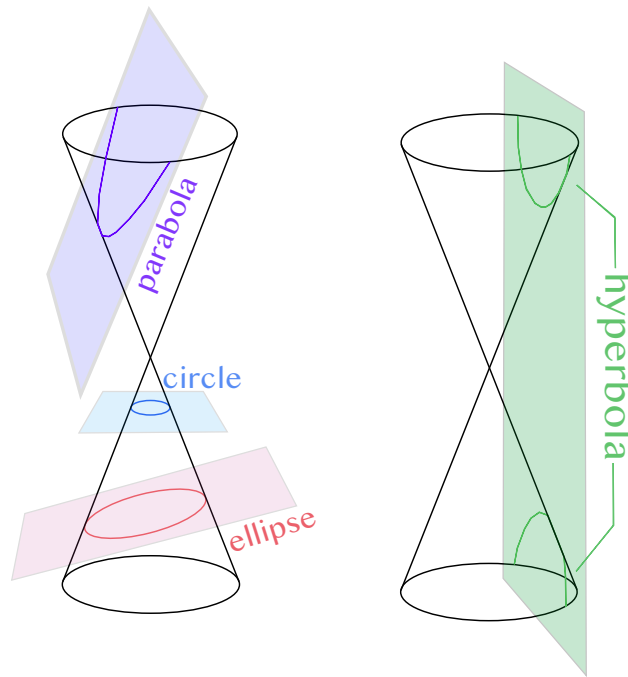


Figure 5: All four types of conic sections

## 2.2 Eccentricity

All of the conic sections are also connected by the idea of **eccentricity**. The eccentricity of a conic section is a quantity that describes how much it “deviates” from a circle.

An eccentricity of 0 means the figure is a circle. (A circle doesn’t deviate at all from being a circle.) Between 0 and 1, the figure is an ellipse. When the eccentricity equals 1, the figure is a parabola, and when the eccentricity is greater than 1, the figure is a hyperbola.

Given a focus and directrix (a point and a line), eccentricity is defined to be the ratio between the distance from any point on the conic section to the focus and the distance from that point to the directrix. So, any conic section can be formed given a focus, directrix, and eccentricity.

## 2.3 General Form

The general form for all conic sections is:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

From this equation we can see that, generally, any polynomial with degree less than or equal to 2 (in  $x$  or  $y$ ) is a conic section. Certain cases for the coefficients of this equation will yield different conic sections, as we will discuss in later sections.

## 3 Circles

The most well-known definition of a circle is its locus definition: the set of all points equidistant from a given point (the center) on a two-dimensional plane. How does this line up with our

definition of conic sections?

We've defined an eccentricity of 0 to be a circle. This also means, by the focus-directrix definition of eccentricity, the ratio of the distance from a point on the circle to some focus and the distance from a point on the circle to some directrix is 0. So where is the focus and directrix for any circle?

Well, if we considered some line  $l$  and a point  $P$ , where would the ratio of those distances be 0? Let  $X$  be any point that can be on our circle. Looking at our definition for eccentricity,

$$e = \frac{d(X, P)}{d(X, l)}$$

The only way  $e = 0$  is possible is when  $d(X, P) = 0$ . The only point  $X$  such that  $d(X, P) = 0$  is  $P$ , so that means our circle is only the point  $P$ ! This doesn't quite work with what we already know about circles, so what's wrong?

The issue is the directrix. As long as  $d(X, l)$  is finite, the figure will be a degenerate circle. So, let's consider,

$$\lim_{d(X, l) \rightarrow \infty} e = \lim_{d(X, l) \rightarrow \infty} \frac{d(X, P)}{d(X, l)} = 0.$$

This is true as long as  $d(X, P)$  is finite. So when the directrix is a line infinitely far away from the circle, all points that are a finite distance from the focus are on the circle? That isn't right either, which is why we need a radius.  $P$ , the focus, is the center of the circle, so a radius would just be a constant  $d(X, P)$ .

So for a non-degenerate circle, the directrix is a line at infinity and the focus is the center of the circle. The directrix ends up being insignificant in this case, because it would be a line at infinity for any circle, so it doesn't help at all. A center and a radius identifies a circle, as we have eccentricity ( $e = 0$ ), a focus (the center), and a directrix (insignificant). The radius is a byproduct of the fact that we can't determine anything about  $d(X, P)$  when the directrix is at infinity except that it is finite.

### 3.1 General Form

In the general form for conic sections, the figure is a circle if  $A = C$  and  $B = 0$ . This can be seen by experimentation, for example looking at  $2(x - 5)^2 + 2(y - 6)^2 = 25$ , we can see that it is a circle centered at (5,6) with radius  $\frac{5}{\sqrt{2}}$ , and it has  $A = C$  and  $D, E$ , and  $F$  terms. However, there is no  $B$  term, which makes sense because an  $xy$  term would come from some form of  $(ax + by)^2$  which would not result in a circle.

## 4 Ellipses

Ellipses are often described as squished circles, because that is exactly what they are. If you take a circle, grab it at diametrically opposite points, and pull on it, you have created an ellipse.

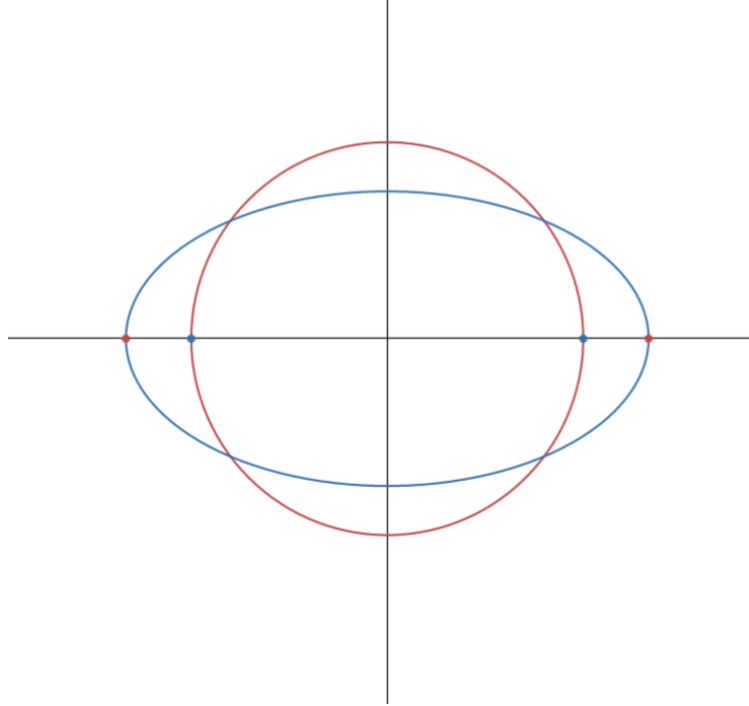


Figure 6: Circle pulled from blue points to red points forms ellipse.

#### 4.1 Foci

An ellipse is most commonly defined as the set of points such that the sum of the distance from two points to a point on the figure is constant. With this definition, an ellipse has two foci, and the sum of the distances from each foci to a point on the ellipse is a constant value. Here we can see the similarity to a circle: A circle is the set of points a constant distance away from the center (a focus), and an ellipse is the set of points such that the distances from its two foci sum to a constant value, which we call the **focal sum**. With our pulling on the circle example, we can imagine the focus of the circle gets pulled apart into two foci. The ellipse still has a center, though, and it's at the midpoint of the line segment connecting the foci.

#### 4.2 Anatomy

The general form for a horizontally-oriented ellipse centered at  $(h, k)$  is:

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

If  $a > b$ , then  $a$  and  $b$  are called the **semimajor** and **semiminor** axes of the ellipse, which are each half of the **major** and **minor** axes of the ellipse. The major axis can be thought of as the line connecting the points farthest apart on the ellipse, and these points are called the **vertices** of the ellipse. The minor axis is the line connecting the points on the ellipse closest to each other. In an ellipse centered at the origin, like in Figure 6,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

The minor axis lies on the  $y$ -axis, and the major axis lies on the  $x$ -axis, as long as  $a > b$ . If  $b > a$ , then the ellipse is vertically-oriented and the major and minor axes are on the  $y$ -axis and

$x$ -axis respectively. In Figure 6, the red points are the vertices of the ellipse. The foci in an ellipse of this form can be found using  $a$  and  $b$ , utilizing the focal sum.

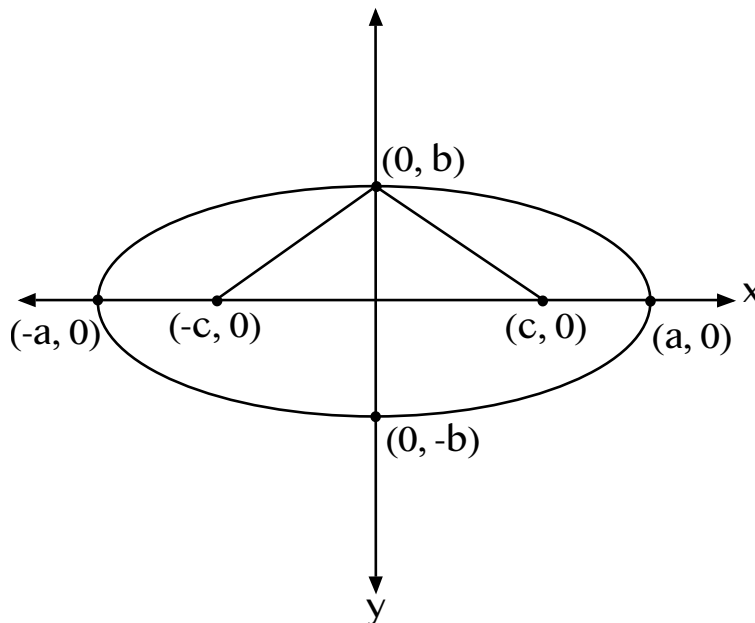


Figure 7: Anatomy of ellipse

The distance from the center of the ellipse to either focus is  $c$ . We can find  $c$  as follows:

- The focal sum to the point  $(a, 0)$  is  $(a - c) + (a + c) = 2a$ .
- Consider the triangle formed by  $(0, b)$ ,  $(-c, 0)$  and  $(c, 0)$ . This triangle is isosceles since the length of the line segments connecting  $(0, b)$  to the foci can both be expressed as  $\sqrt{c^2 + b^2}$ .
- The focal sum  $2a$  gives us  $2a = 2\sqrt{c^2 + b^2}$ , which yields  $a^2 = b^2 + c^2$ .

So now, given an ellipse in its general form, we are able to find its foci.

### 4.3 Focus-Directrix

For an ellipse, the eccentricity is between 0 and 1. To say this, there must be some way an ellipse can be defined with a focus and directrix. And there is! For a horizontally oriented ellipse centered at the origin, consider some vertical line  $x = k$ . We know that this line can't intersect the ellipse, so let it be to the right of the ellipse.

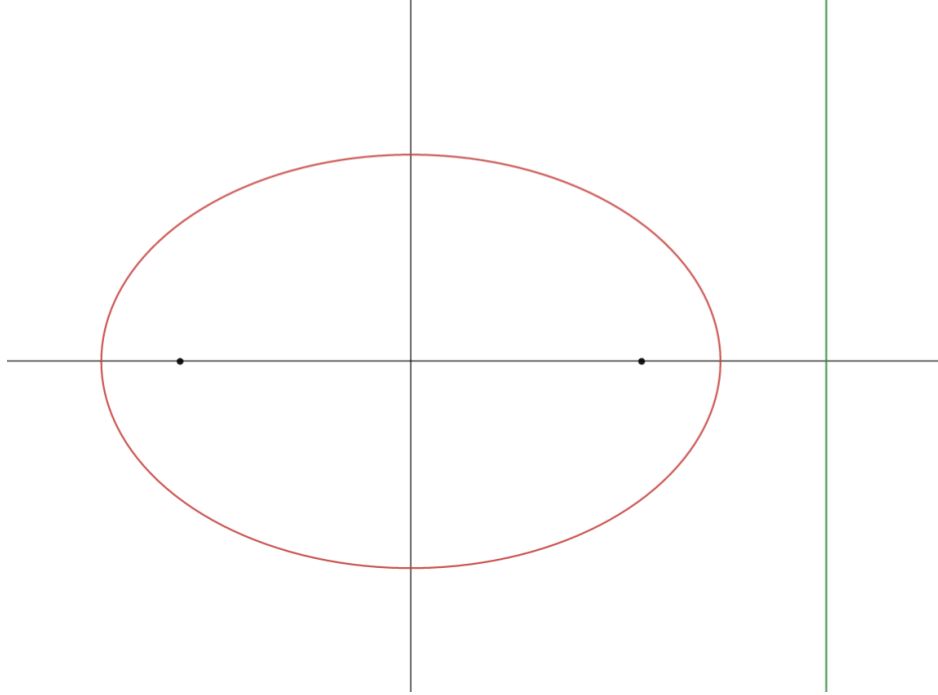


Figure 8: Ellipse, with foci at black points and line  $x = k$

We know the ratio of the distance from a point on the ellipse to the focus to the distance from that point on the ellipse to the directrix is the eccentricity. Consider  $(a, 0)$ , the vertex of the ellipse on the right. The distance from this vertex to the focus is  $a - c$ , and the distance from this focus to the directrix is  $k - a$ . We can write

$$e = \frac{a - c}{k - a}$$

Now, consider the eccentricity at  $(-a, 0)$ :

$$e = \frac{a + c}{k + a}$$

These values are equivalent as they are both the eccentricity of the ellipse, so we can solve for  $k$ :

$$\begin{aligned} \frac{a + c}{k + a} &= \frac{a - c}{k - a} \\ (a + c)(k - a) &= (a - c)(k + a) \\ ak - a^2 + ck - ac &= ak + a^2 - ck - ac \\ 2ck &= 2a^2 \\ k &= \frac{a^2}{c} \end{aligned}$$

Now we have found the equation for the directrix in terms of  $a$  and  $c$ .

#### 4.4 Eccentricity, Part 2

Eccentricity can also be defined based on the conic section itself. For a circle, it's 0, and for a parabola, it's 1. For an ellipse, however, the eccentricity is always between 0 and 1, and can be



defined as

$$e = \frac{c}{a},$$

where  $a$  is the semimajor axis and  $c$  is the distance from the center to a focus. We can prove this new definition for eccentricity using our equation for the directrix. Consider  $(a, 0)$  again: The distance from  $(a, 0)$  to the focus is  $a - c$ , and the distance from  $(a, 0)$  to the directrix is  $\frac{a^2}{c} - a$ . Simplifying, we get:

$$\begin{aligned}\frac{a - c}{\frac{a^2}{c} - a} &= \frac{ac - c^2}{a^2 - ac} \\ \frac{ac - c^2}{a^2 - ac} &= \frac{c(a - c)}{a(a - c)} \\ e &= \frac{c}{a}\end{aligned}$$

This makes sense in the context of the “pulling on a circle” example: Since the focus of the circle is also the center,  $c = 0$ , so the eccentricity of a circle is 0. Since  $0 < \frac{c}{a} < 1$ ,  $c < a$ . So, as long as  $c < a$ , or the semimajor axis is greater than the distance from the focus to the center,  $e < 1$ , so the figure is an ellipse. If  $c > a$ , then the focus is forced outside of the ellipse, and this becomes a hyperbola and  $e > 1$  (although  $c$  is defined differently for hyperbolas, which will be discussed later). Notice that our derivation breaks when  $c = a$ , which is when the figure will be a parabola, not an ellipse (it doesn’t make sense for the focus to be on a vertex of the ellipse).

We can also express the equation for the directrix in terms of eccentricity.

$$k = \frac{a^2}{c}$$

We can simplify this with  $e = \frac{c}{a}$ :

$$k = \frac{a}{e}$$

Now we’ve shown that the directrix of a horizontally-oriented ellipse centered at the origin is at  $x = \frac{a}{e}$ . Note that this allows us to use one focus with the directrix to form the ellipse.

## 4.5 General Form

In the general form for a conic, the figure will be an ellipse if  $A \neq C$  and  $B = 0$ . This makes sense since an ellipse is a circle if  $a = b$ , when the coefficients of  $x^2$  and  $y^2$  are equal, but when  $a \neq b$ , the figure is an ellipse. Ellipses can, however, have a nonzero  $B$  term, and this is the case in ellipses that are not horizontally or vertically oriented.

## 5 Parabolas

As mentioned in the introduction, parabolas are often used for estimating projectile motion. This is because, when assuming the acceleration of an object is constant (usually from gravity), the function for position of that object will be a quadratic. The general form for a parabola is

$$y = ax^2 + bx + c.$$

The focus-directrix definition for a parabola is fairly well known as the set of points equidistant from a focus and directrix. From this definition, it is easy to see that the eccentricity for parabolas is 1. The standard form for a parabola arises from this definition, which is

$$4p(y - k) = (x - h)^2,$$

where  $p$  is the distance from the vertex to the directrix (or the focus). Both of these formulas are for vertically-oriented parabolas; horizontally-oriented parabolas can be obtained by switching  $(x - h)$  and  $(y - k)$ .

## 5.1 Eccentricity

Let's make more sense of our previous definition of eccentricity ( $e = \frac{c}{a}$ ) for a parabola. Consider an ellipse with a fixed focus  $F$  and a directrix  $l$  to the right of the focus. Right now,  $0 < e < 1$ , and we want  $e$  to approach 1. We will do this by increasing  $c$  and  $a$  by some amount  $x$ , which can be represented by moving the center of the ellipse  $x$  units. Notice that, when moving the center, the other focus of the ellipse also changes such that it is  $c$  away from the new center. Consider

$$\lim_{x \rightarrow \infty} \frac{a + x}{c + x} = 1$$

This is true since  $a$  and  $c$  in this equation are constant, and we increase it by some value  $x$ . This can be visualized with the following figures:

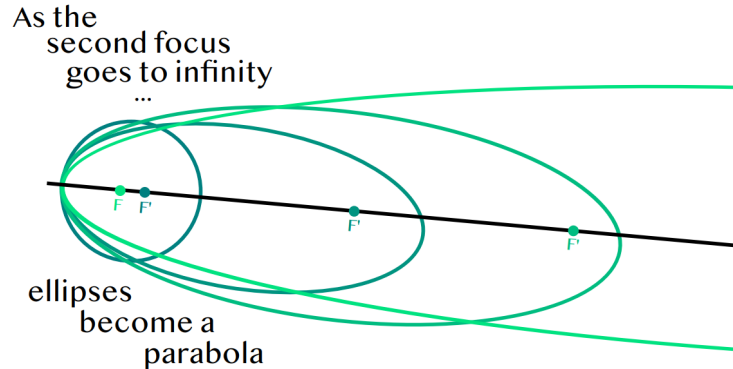


Figure 9: Parabola from ellipse

It can be observed that a parabola is essentially an ellipse with a second focus “at infinity.” In the case of projectile motion on Earth, as seen in the introduction, the Earth’s center is so far away from the ball’s path on the Earth’s surface, so instead of needing to model the path with an ellipse, it can be estimated with a parabola.

## 5.2 General Form

Recall the General form for a conic section:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

For a parabola, in the general form of a conic, if  $B = 0$ , either  $A = 0$  or  $C = 0$ , so there can only be one squared term. We can also have a nonzero  $B$  term with a parabola if we rotate the

figure, just like with an ellipse. If  $B = 0$  and there is an  $x^2$  term, then there must be a  $y$  term to be a parabola. Otherwise, the figure would be two vertical lines. For example,  $x^2 + 1 = 0$  will be the lines  $x = -1$  and  $x = 1$ , but  $x^2 - y = 0$  is just the parabola  $y = x^2$ . We have to set a restriction on the  $B$  term since the  $B$  term causes weird stuff to happen, for example something like  $x^2 + xy + 1 = 0$  ends up being a hyperbola.

### 5.3 The $xy$ term

The  $B$  term of the general form for conics creates a few issues when trying to categorize conics based on coefficients of the terms. Notice that we don't usually have this term in general forms for specific conics such as  $y = ax^2 + bx + c$  or  $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ , as the  $x$  and  $y$  terms are usually separate. So where does the  $xy$  term come from? I've mentioned that there can be a  $B$  term if the figure is rotated, or just not specifically vertically or horizontally oriented.

Let's see this with an example: Take the ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$ :

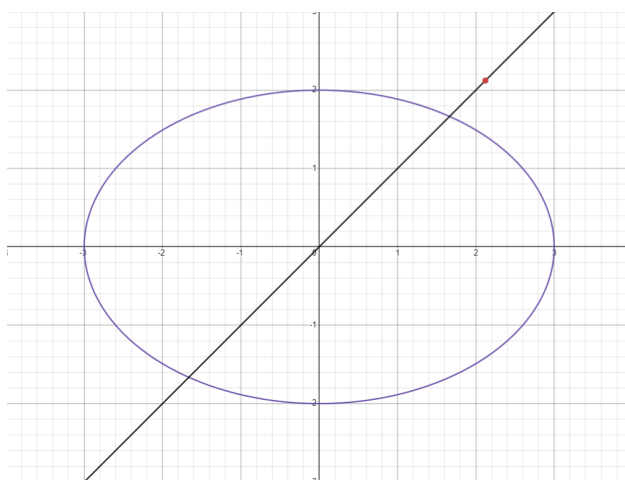


Figure 10:  $\frac{x^2}{9} + \frac{y^2}{4} = 1$

Let's rotate the ellipse  $45^\circ$  counterclockwise. Looking at the point  $(3,0)$ , since rotation is a rigid motion, after rotating this point should still be 3 units away from the origin. Instead of forming an angle of  $0^\circ$  with the x-axis, however, after rotation, the point would lie on a line that creates a  $45^\circ$  angle with the origin. We can find the coordinates of this point by considering an isosceles right triangle with hypotenuse 3, which will have a side length of  $\frac{3}{\sqrt{2}}$ . So, by rotating by  $45^\circ$ , we have transformed the point  $(3,0)$  to  $(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}})$ . Obviously I chose the point and angle so that the rotation would be nice, but this can be generalized. We can consider a rotation as doing a 'change of variables' that rotates the  $xy$  plane to some  $uv$  plane about the origin by some angle  $\theta$ , and the following relation is true:

$$\begin{aligned} u &= x \cos \theta - y \sin \theta \\ v &= y \cos \theta + x \sin \theta \end{aligned}$$

We can see that  $\frac{3}{\sqrt{2}} = 3 \cos 45^\circ - 0 \sin 45^\circ$  and  $\frac{3}{\sqrt{2}} = 0 \cos 45^\circ + 3 \sin 45^\circ$ , so this works for our example. Just for fun, let's prove this:

Dealing with rotation in the rectangular plane is annoying, so let's use polar coordinates.

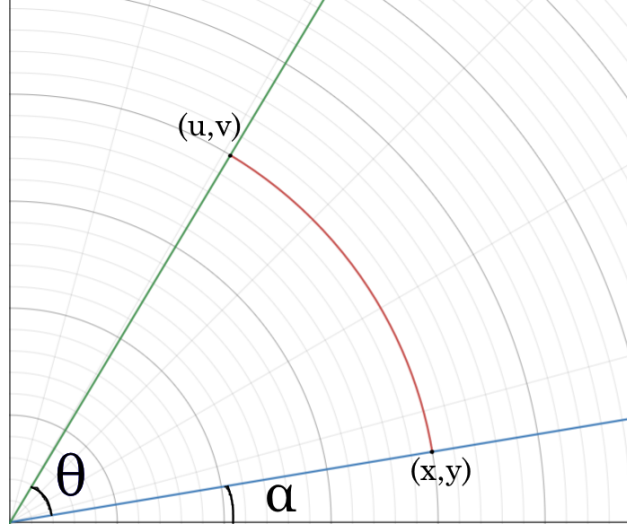


Figure 11: Polar Coordinate System (The points are in rectangular form)

Since we are rotating about the origin, the distance from  $(0,0)$  to  $(x,y)$  is the same as the distance from  $(0,0)$  to  $(u,v)$ . Let  $r$  denote that common distance. Denoting  $\alpha$  and  $\theta$  as they are on Figure 11, we have the following equations:

$$x = r \cos \alpha \quad (1)$$

$$y = r \sin \alpha \quad (2)$$

$$u = r \cos(\alpha + \theta) \quad (3)$$

$$v = r \sin(\alpha + \theta) \quad (4)$$

We can apply the sin and cos addition formulas to (3) and (4) to get:

$$u = r \cos \alpha \cos \theta - r \sin \alpha \sin \theta$$

$$v = r \sin \alpha \cos \theta + r \sin \theta \cos \alpha$$

Plugging in (1) and (2), we get:

$$u = x \cos \theta - y \sin \theta$$

$$v = y \cos \theta + x \sin \theta$$

So we have shown that we can rotate the  $xy$  plane to some  $uv$  plane by any angle  $\theta$  using these formulas. Let's see what happens when we use this with the ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$  rotated by  $45^\circ$ . Our rotated ellipse is only rotated in the  $xy$  plane; it is vertically oriented in the  $uv$  plane (since the  $uv$  plane is a rotation of the  $xy$  plane). So, in the  $uv$  plane, we are dealing with  $\frac{u^2}{9} + \frac{v^2}{4} = 1$ ,

and we can substitute the above equations to find this in the  $xy$  plane:

$$\begin{aligned}
\frac{u^2}{9} + \frac{v^2}{4} &= 1 \\
u &= x \cos \theta - y \sin \theta \\
v &= y \cos \theta + x \sin \theta \\
\frac{(x \cos \theta - y \sin \theta)^2}{9} + \frac{(y \cos \theta + x \sin \theta)^2}{4} &= 1 \\
\frac{x^2(\cos \theta)^2 - 2xy \cos \theta \sin \theta + y^2(\sin \theta)^2}{9} + \frac{y^2(\cos \theta)^2 + 2xy \cos \theta \sin \theta + x^2(\sin \theta)^2}{4} &= 1 \\
4x^2(\cos \theta)^2 - 8xy \cos \theta \sin \theta + 4y^2(\sin \theta)^2 + 9y^2(\cos \theta)^2 + 18xy \cos \theta \sin \theta + 9x^2(\sin \theta)^2 &= 36 \\
4x^2 \cos^2 \theta + 9x^2 \sin^2 \theta + 4y^2 \sin^2 \theta + 9y^2 \cos^2 \theta + 10xy \cos \theta \sin \theta &= 36 \\
4x^2((\cos \theta)^2 + (\sin \theta)^2) + 5x^2(\sin \theta)^2 + 4y^2((\sin \theta)^2 + (\cos \theta)^2) + 5y^2(\cos \theta)^2 + 10xy \cos \theta \sin \theta &= 36 \\
4x^2 + 4y^2 + 5x^2(\sin \theta)^2 + 10xy \cos \theta \sin \theta + 5y^2(\cos \theta)^2 &= 36
\end{aligned}$$

Or, expressed in the general form:

$$(4 + 5(\sin \theta)^2)x^2 + (4 + 5(\cos \theta)^2)y^2 + (10 \cos \theta \sin \theta)xy - 36 = 0$$

Now, finally, let's plug in  $45^\circ$ :

$$\begin{aligned}
(4 + 5(\sin 45^\circ)^2)x^2 + (4 + 5(\cos 45^\circ)^2)y^2 + (10 \cos 45^\circ \sin 45^\circ)xy - 36 &= 0 \\
(4 + \frac{5}{2})x^2 + (4 + \frac{5}{2})y^2 + (\frac{10}{2})xy - 36 &= 0 \\
\frac{9}{2}x^2 + \frac{9}{2}y^2 + 5xy - 36 &= 0
\end{aligned}$$

As we can see, rotated conic sections create these  $xy$  terms (and are often pretty ugly too). This is why it becomes difficult to generalize conics by the general form, since the equations for any specific conic can take many different forms. We can also see here that  $A = C$ , so we need a restriction on  $B$  when making the claim in 4.2.

## 6 Hyperbolas

Hyperbolas are the least appreciated conic section. Circles and parabolas are the most known, and ellipses are fairly well known, but hyperbolas are more obscure. Hyperbolas can be thought of as “inverted” ellipses.

Hyperbolas have a very similar definition to ellipses. An ellipse is the set of points such that the sum of the distances from the foci are a constant value, the focal sum. A hyperbola is the set of points such that the absolute value of the difference of the distances from two points is constant. From this definition, the general form for a hyperbola emerges:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

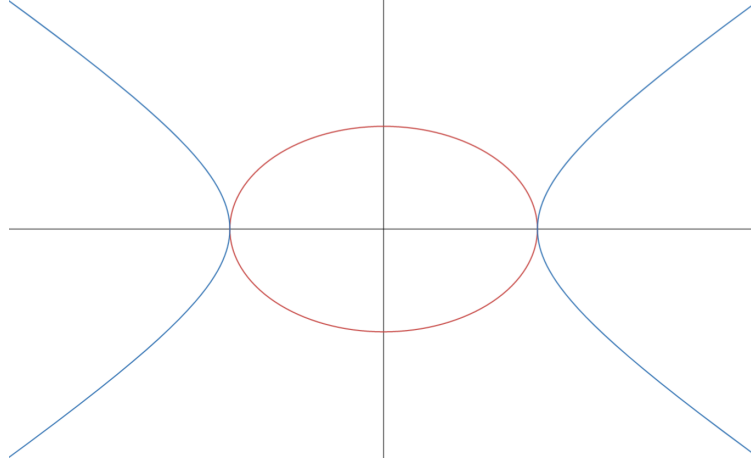


Figure 12:  $\frac{x^2}{9} + \frac{y^2}{4} = 1$  in red,  $\frac{x^2}{9} - \frac{y^2}{4} = 1$  in blue

## 6.1 Anatomy

The anatomy of a hyperbola is similar to an ellipse as well, with some key differences. Figure 12 is a diagram with the equations  $\frac{x^2}{9} + \frac{y^2}{4} = 1$  and  $\frac{x^2}{9} - \frac{y^2}{4} = 1$ .

Notice that  $a$  and  $b$  are the same for the ellipse and the hyperbola. The major axis of the ellipse is also called the transverse axis of the hyperbola, and the minor axis of the ellipse is also called the conjugate axis of the hyperbola. The major, minor, transverse, and conjugate axes are all axes of symmetry for each figure. They have the same vertices, but the foci are at different places, and thus  $e$  is a different value.

The value of  $c$ , the distance from the center to the focus, is different in hyperbolas and ellipses with relation to  $a$  and  $b$ . We showed that  $a^2 = b^2 + c^2$  or  $c^2 = b^2 - a^2$  for ellipses, and it can be similarly shown that  $c^2 = a^2 + b^2$  for hyperbolas. Now it makes sense that eccentricity is greater than 1 for hyperbolas, since  $e = \frac{c}{a} = \frac{\sqrt{a^2+b^2}}{a} = \sqrt{\frac{a^2+b^2}{a^2}} = \sqrt{1 + \frac{b^2}{a^2}} > 1$ .

We can now use the anatomy to derive the general form for a hyperbola. By definition,  $|d_2 - d_1|$  is constant. Consider the focal difference at point  $(a, 0)$  (Refer to Figure 13):

- The focal difference to the point  $(a, 0)$  is  $(2a + c) - (c) = 2a$ .
- By the pythagorean theorem,  $d_2 = \sqrt{y^2 + (c + x)^2}$  and  $d_1 = \sqrt{y^2 + (x - c)^2}$
- We now have  $\sqrt{y^2 + (c + x)^2} - \sqrt{y^2 + (x - c)^2} = 2a$ .
- Simplifying, we get  $x^2(c^2 - a^2) - a^2y^2 = a^2(c^2 - a^2)$
- Plugging in  $b^2 = c^2 - a^2$ , we get the general form for a hyperbola:

$$b^2x^2 - a^2y^2 = a^2b^2$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Just as parabolas can be thought of as ellipses that “break” as the focus goes to infinity, hyperbolas can be thought of as ellipses that go “around” infinity, with the second focus of the ellipse coming back from negative infinity.

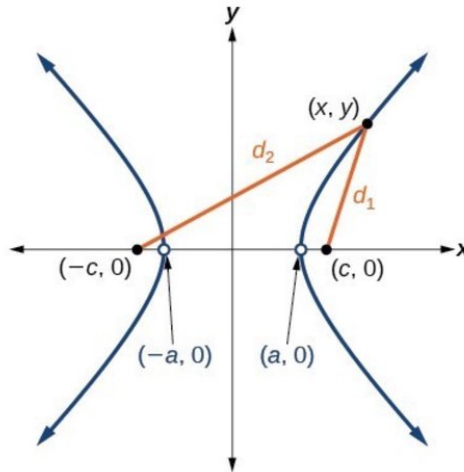


Figure 13: Hyperbola anatomy

## 6.2 Linear Asymptotes

One cool aspect of hyperbolas are that they have linear asymptotes. We can see this by manipulating the general form:

$$\begin{aligned}
 \frac{x^2}{a^2} - \frac{y^2}{b^2} &= 1 \\
 \frac{y^2}{b^2} &= \frac{x^2}{a^2} - 1 \\
 y^2 &= \frac{b^2 x^2}{a^2} - b^2 \\
 y^2 &= \frac{b^2}{a^2} (x^2 - a^2) \\
 y &= \pm \frac{b}{a} \sqrt{x^2 - a^2}
 \end{aligned}$$

To analyze end behavior, we can observe that as  $x$  gets arbitrarily large, it will dominate the square root term, so for large  $x$ , this is approximately:

$$\begin{aligned}
 y &= \pm \frac{b}{a} \sqrt{x^2} \\
 y &= \pm \frac{b}{a} x.
 \end{aligned}$$

With this result we have shown that as  $x$  approaches infinity, hyperbolas asymptotically approach the lines  $y = \pm \frac{b}{a} x$ .

## 7 Real World Applications

You might be thinking, okay, all conics are connected, so what? Looking past the math of conic sections, they have applicable uses in real life. I've already mentioned projectile motion, but it is very useful to make the approximation from an ellipse to parabola for ease of calculation.

## 7.1 Orbits

Newton's laws of universal gravitation can be used to show that the path of all orbits are conic sections. Kepler's first law also states that all planets move in elliptical orbits. For example, the Earth's orbit around the sun and the Moon's orbit around the Earth are elliptical. Satellite orbit around the Earth can be either circular or elliptical. Parabolic and hyperbolic paths occur when the object in orbit is at or past escape velocity. When the object is exactly at escape velocity, the path is parabolic. If the object is travelling faster than escape velocity, like a comet, it will follow a hyperbolic path. The linear asymptotes of a hyperbola represent the path of the comet as it becomes arbitrarily far away from the sun; as the sun has less and less gravitational pull on it, the comet will travel in a straight line (as long as no other forces act on it).

## 7.2 Reflective Properties

Each conic section has its own reflective property that makes it useful to us. I won't prove them here, but feel free to try them on your own.

Parabolas have the property that any line originating from the focus will be reflected at a path perpendicular to the directrix. The opposite is true as well, so this is useful for sending out signals as well as collecting signals. Paraboloids are 3D objects that are just solids of revolutions of a parabola. These are used in real life for satellite dishes and car headlights. This property makes paraboloid shapes very useful since they will work even if a signal is very weak, as long as the signal is pointed towards the dish.



Figure 14: Satellite dish collecting signal at focus of circular paraboloid

Ellipses have the property that any path originating from one focus will be reflected off of the ellipse to the other focus. We can see the connection with the parabola reflective property here, since the parabola's second focus is essentially at infinity, so the reflected lines are parallel lines which meet "at infinity." Furthermore, signals sent from one focus at the same time will reach the second focus at the same time. This, again, is used to send signals, as well as some obscure areas such as lithotripsy.

Hyperbolas have the property that any path directed at one focus will be deflected by the hyperbola towards the other focus. This is similar to the parabola in that it can take paths pointed at one place and redirect it to another. This is used in radio direction (between two towers) and is



useful for mirrors in telescopes to reflect light towards the eyepiece.

### 7.3 Hyperbolic Geometry

Hyperbolas also have interesting connections to non-Euclidean geometry. A figure called the hyperbolic paraboloid was one of the first used to investigate non-Euclidean geometric properties. It also looks like a potato chip(see Figure 15).

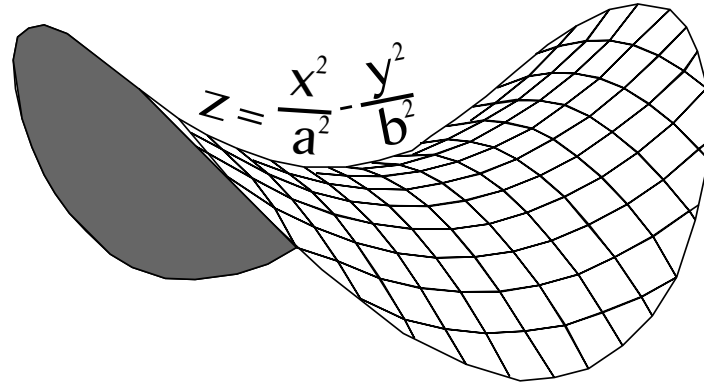


Figure 15: Hyperbolic paraboloid

Hyperbolic geometry is very useful in the studies of special and general relativity and descriptions of spacetime. It also has many other interesting practical uses, such as modelling human vision or in cameras.

## 8 Conclusion

Conic sections may just seem like some obscure topic in your math class, but they can essentially be looked at as the curves of nature. They can be used to describe the very fabric of our universe and for our own technological advances. Conics may be challenging to work with mathematically, but seeing how they are all connected through eccentricity and of all things, a cone, makes it a fascinating subject.